lift-off

FRAALLIANCE CHALLENGE

30 October 2026

walk-forward-v6

48-hour runway forecast

MODEL CARD / FRANKFURT AIRPORT

PURPOSE AND MODEL

Hourly forecasts up to 48 h ahead: operating direction 25/07, probability of runway-18 weather usability, certainty and next likely change. Two gradient-boosted classifiers use 45 features and persistence blending. Intended for planning support; actual operating decisions remain with DFS.

PUBLIC DATA AND KEY DRIVERS

DFLD supplies half-hour operating-direction records from 2018; IEM supplies EDDF METAR. Open-Meteo Previous Runs supplies archived ECMWF, GFS and ICON weather forecasts from 2024. The FraAlliance brief defines the rule baseline. Runway-18 targets are inferred from METAR, not observed closures.

Top permutation drivers in the final test month: north-south pressure difference, current direction, ensemble tailwind on 25, ECMWF tailwind on 25 and pressure-derived tailwind.

EVALUATION AND QUALITY

Twelve expanding monthly folds, November 2025-October 2026; training targets end before each test month. Issues every 6 h; archived forecast availability uses a six-hour publication proxy. Stable-hour accuracy on matched labelled rows:

Lead	Model	Base-wind rules	Keep-current	Rows
1 h	**95.7%**	93.1%	95.9%	1,416
6 h	**92.7%**	88.1%	88.8%	1,416
24 h	**91.8%**	87.7%	75.1%	1,383
48 h	**88.9%**	84.0%	64.2%	1,338

24 h Brier score: model 0.0637; rules 0.1229. At 6/24/48 h, seven-day block-bootstrap 95% intervals for the accuracy advantage over rules are +2.7 to +6.5 / +3.1 to +6.3 / +3.5 to +7.2 percentage points. No accuracy advantage over persistence is established at 1 h.

LIMITS

The folds informed feature selection and review; they are not an untouched final test. Exact weather publication times, DFLD publication delays and summer-clock alignment remain unverified. DFLD is a public proxy; transition hours are excluded from the accuracy table. Certainty is a probability-margin score and the change window is heuristic, neither calibrated.

Runway-18 skill is unproven: at 24 h, 0 of 12 positive windows detected, 0 false alarms, 307 complete windows; 1,125 excluded. These are overlapping windows with inferred weather labels, not unique operational closures.

Evidence and reproduction: Technical card, published results, submission and run commands.